INCH-POUND

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PERFORMANCE SPECIFICATION

TRANSMITTER RATE OF FLOW, IN-FLIGHT REFUELING, 0-10,000 PPM, TYPE MB-3

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This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of in-flight refueling rate of flow and fuel transmitter, designated Type MB-3.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-38120	- Indicator, Rate of Flow and Transfer EFU-20/A

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

(Copies of these documents are available online at http://quicksearch.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM-B117	- Salt Spray (Fog) Testing Apparatus, Operating
ASTM-D910	- Gasolines, Aviation

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(ASTM documents may be obtained online at http://www.astm.org/ or from American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE-ARP8615	- Fuel System Components: General Specification for
SAE-AS31061	- Connector, Plug, Electric, Straight, Solder Contacts AN Type
SAE-AS31081	- Connectors, Plug, Electric, Solder Contact, 90 Degree, AN Type

(SAE documents may be obtained online at http://www.sae.org/ or from SAE International Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 USA.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1 and 4.3.

3.2 Function. The transmitter shall be designed to provide an electrical output signal for a fuel flow range of 0 to 10,000 pounds per minute (ppm). The signal shall be suitable for remote indication and for use in operating an EFU-20/A indicator conforming to MIL-PRF-38120.

3.3 Design and Construction.

3.3.1 Output signal. The electrical signal shall be single phase, 400 Hertz, AC voltage. The magnitude of the voltage shall be 1 V per 1,00 ppm fuel flow. The electrical output signal shall have a carrier frequency identical to the input power frequency. The output voltage signal shall be at a leading phase angle of 15° with respect to the input voltage.

3.3.2 Excess flow. The transmitter shall not be damaged by fuel flow rates in excess of the rated value provided the pressure rating of the transmitter is not exceeded.

3.3.3 Fuel variations. The transmitter shall be designed to operate satisfactorily when using fuel having any of the variations in standard characteristics specified in MIL-DTL-5624 and ASTM-D910.

3.3.4 Operating pressure. The transmitter shall be designed for a normal operating pressure of 120 psi, a proof pressure of 240 psi, and a burst pressure of 480 psi.

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3.3.5 Electrical power. The electrical power for the output signal circuit shall be 115 V AC, 400 Hertz. There shall be no voltage phase angle difference between this power and the power to the rest of the system. The wave form of the supply voltage may be nonsinusoidal with a crest factor of 1.414 ±10% and with harmonics having respective amplitudes that are less than 5% of the fundamental.

3.3.6 External wiring. The in-flight refueling system interconnecting wiring diagram shall conform to Figure 1.

3.3.7 Density correction. The transmitter shall measure the fuel flow rate on a gravimetric basis without a separate and distinct density correcting means.

3.3.8 Fuel line connections. The transmitter shall be designed for mounting by means of the fuel line connections or by straps. The fuel line connections shall be as specified by the procuring activity and shall be furnished with a suitable plugs or caps to prevent contamination by foreign material during storage and shipment. The fuel system tubing adjacent to the transmitter shall be in accordance with requirements of the procuring activity.

3.3.9 Damping. The transmitter shall be suitably damped.

3.3.10 Dimensions. The dimensions of the transmitter shall conform to Figure 2.

3.3.11 Flow direction. The direction of fuel flow through the transmitter shall be permanently and legibly marked on each side by an arrow as shown on Figure 2.

3.3.12 Weight. The weight of the transmitter shall not exceed 18.0 pounds.

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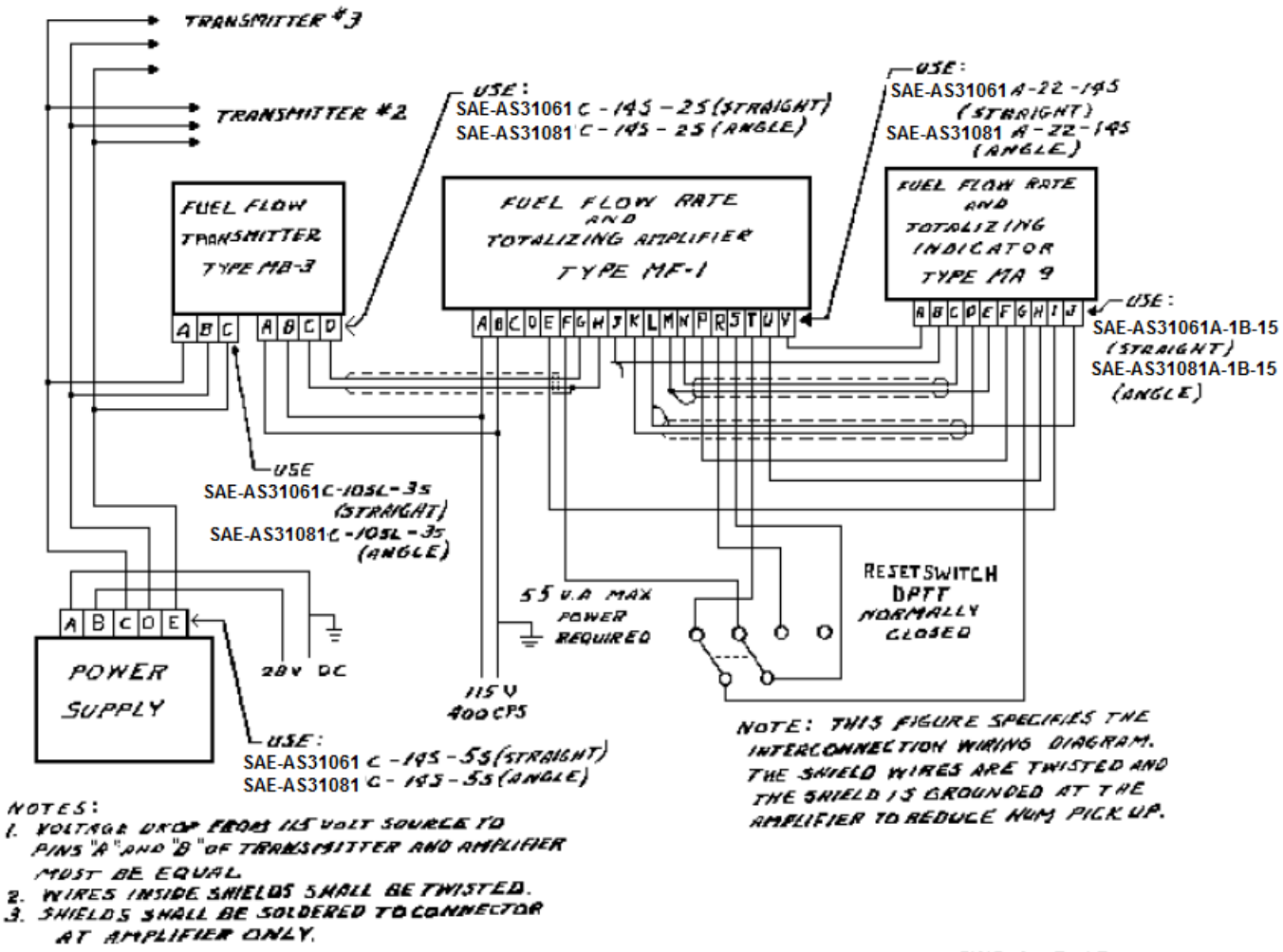
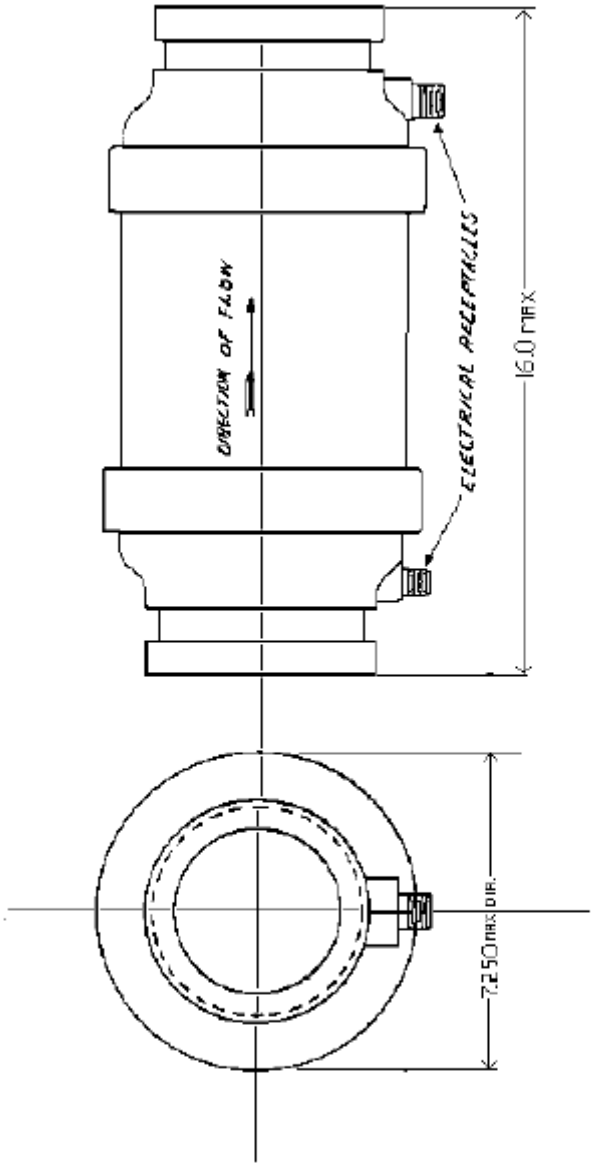


FIGURE 1 INFLIGHT REFUELING SYSTEM-INTERCONNECTING WIRING.

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Notes:
1. Dimensions are in inches.

FIGURE 2. Transmitter.

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3.4 Performance. The transmitter shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Temperature - Temperatures within the range of -54° to +71°C.

b. Humidity - Relative humidity up to 100%.

c. Salt spray - Exposure to salt sea atmosphere for a 50 hour period.

d. Vibration - Vibration incident to service use.

e. Fungus - Fungus growth as encountered in tropical climates.

f. Explosion proof - Operation in an explosive vapor within or surrounding the equipment.

g. Altitude - Pressures ranging from 30 inches Hg down to 3.4 inches Hg (approximately an altitude of 50,000 feet).

h. Shock - Shock forces as anticipated.

i. Dielectric strength - A potential of 550 V AC, 60 Hertz for a period of 10 seconds when applied as specified in 4.5.

j. Leakage - A hydraulic proof pressure of 240 psi applied and maintained for 5 minutes.

k. Cycling operation - The system shall be capable of operating for 500 hours of actual or simulated flow through the transmitter without failure when tested in accordance with 4.5.26.

3.4.1 Radio noise interference. The transmitter shall meet requirements of MIL-STD-461, Emission and Susceptibility Requirements CE102 and RE102, except that the frequency range shall be 0.15 to 150 MHz.

3.4.2 Contaminated fuel operation. The transmitter shall operate satisfactorily with fuels contaminated to the extent specified in 4.5.28.

3.5 Materials. Nonmagnetic and nonferrous materials shall be used for all parts of the transmitter, except where such materials are essential. Metals shall be corrosion resistant or suitably treated to resist corrosion. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with one another.

3.5.1 Protective treatment. When materials are used in the transmitter that will deteriorate when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that

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will crack, chip, or scale with age or the extremes of climatic and environmental conditions shall be avoided.

3.5.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.6 Identification plate. An identification plate shall be provided on transmitter. Information provided shall include the government designation for the transmitter, the manufacturer's part number and the manufacturer's Commercial and Government Entity (CAGE) Code.

3.7 Workmanship. The transmitter shall be constructed and finished in a thoroughly workmanlike manner so that no parts will work loose service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, welding, riveting, brazing, plating, machine screw assemblies, impregnation of coils, and painting. All parts shall be free from burrs and sharp edges. Gears shall be of high quality and shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots or other irregularities.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.1.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.2.

4.2 Test conditions.

4.2.1 Atmosphere. Whenever the pressure and temperature existing at the time of the test are not otherwise specified, the test shall be made at atmospheric pressure (29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from specified condition.

4.2.2 Tapping. Unless otherwise specified, the transmitter shall be gently tapped or vibrated before a test reading is taken.

4.2.3 Test position. Unless otherwise specified, the transmitter shall be tested in the normal operating position. The transmitter will be in the normal operating position when

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horizontal with the electrical connectors at 3 o'clock position when viewed from the inlet end of the transmitter.

4.2.4 Test power. The test power of the transmitter shall be as follows:

a. The 3 phase power into the transmitter shall be 400 cycles. This frequency shall be held by the power supply to within $\pm$0.1%. The AC voltage fed into the power supply inverter shall be 28 ±0.5 VDC.

b. The voltage input to the output signal network shall be 115 ±0.1%V AC with a frequency of 400 ±2 Hertz.

c. The 400 Hertz supply voltage may be nonsinusoidal with a crest factor of 1.414 ±10% and with harmonics whose respective amplitudes are less than 5% of the fundamental.

4.2.5 Test fuel. The fuel in testing or for calibration purposes shall conform to MIL-DTL-5624 (JP-4). The specific gravity of the fuel shall be 0.785 ±0.005 at room temperature. A commercial equivalent with a higher flash point may be used.

4.2.6 Master flowmeter. A master flowmeter to establish flow rate may be used. The calibration of this master shall be verified by the procuring activity.

4.2.7 Master voltmeter. The output signal from the transmitter shall be measured with a master voltmeter accurate to within ±0.1%. The input impedance of the voltmeter shall be 100,000 Ω ±5% resistive load or better.

4.2.8 Phase angle meter. The phase angle meter shall be accurate to $\pm 0.1^{o}$.

4.3 First article. The first article inspection shall consist of all tests described under paragraph 4.5.

4.3.1 Inspection sample. The first article inspection sample shall consist of 3 transmitters which are identical in all respects to the transmitters that are produced as production items.

4.3.2 First article test report. Upon completion of the first article inspection and tests, a test report shall be prepared in accordance with a contractor format. The report shall include the results of the tests conducted as well as checklists used during testing.

4.4 Conformance inspection. Inspection of transmitters shall consist of individual tests and sampling tests.

4.4.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.5.

a. Physical examination

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b. Dielectric strength

c. Leak test

d. Scale error and friction

e. Phase angle tolerance

4.4.2 Sampling tests. Unless otherwise specified a minimum of 2 transmitters shall be selected at random from each 100 items produced on the contract or order and subjected to all the tests as described under 4.5.

4.4.3 Rejection and Retest. When an item selected from a production run fails to meet the tests of this specification, no items still on hand or later produced shall be accepted until the extent and cause of the failure have been determined and corrected. The contractor shall provide a written explanation as to the cause of the failures and the actions taken to preclude recurrence. After correction, all tests shall be repeated. For production reasons individual tests may be continued pending the investigation of a sampling test failure but final acceptance of these items shall not be made until it is determined that all items meet all the requirements of this specification. IF investigation indicates that defects may exist in items already accepted the contractor shall notify the procuring activity of these defects and the method of correcting them.

4.5 Test Methods.

4.5.1 Physical examination. The transmitter shall be inspected to determine compliance with this specification with respect to materials, workmanship, identification and dimensions.

4.5.2 Dielectric strength. Prior to sealing, 550 V AC at a commercial frequency shall be applied to certain pins and the case for a period of 5 seconds. Only those pins not used for grounding the components to the case shall be used. There shall be no insulation breakdown.

4.5.3 Leak test. The transmitter shall be connected to a hydraulic pump or other source of hydraulic pressure and a proof pressure of 240 psi applied and maintained for a period of 5 minutes. There shall be no leakage or damage as a result of this test.

4.5.4 Scale error and friction. The test points shall be as noted in Table I. The scale error shall not exceed the tolerance in Table I. The friction error shall be determined at the mid-scale point by bringing the flow up to, but not exceeding, the specified rate and then down to, but not below, the specified rate. The difference between the upscale and downscale reading shall not exceed the tolerance in Table II. Tapping shall not be used when making the friction error readings. The transmitter shall be tested on a flow stand capable of determining the true flow within an accuracy of ±0.5% of the point.

4.5.5 Phase angle tolerances. The phase angle tolerances at the test points shall not exceed the values shown in Table III.

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4.5.6 Position error. In determining the position error, two additional calibration tests shall be conducted. The first test position shall be the normal test position. The second test position shall be horizontal with the connectors at the 6 o'clock position when looking at the inlet end of the transmitter. The third test position shall be horizontal with connectors at the 9 o'clock position when looking at the transmitter from the inlet end. The calibration errors in positions 2 and 3 shall not exceed the room temperature tolerance in position 1 by more than that specified in Table II.

TABLE I. Scale error tolerance.

Scale Error Tolerance ± PPM					
Flow PPM	Room Temp	Low Temp	High Temp	Substitute Fuel	Cycling Test
0	20	30	30	30	40
1,000	20	30	30	30	40
2,000	25	35	35	35	45
3,000	40	55	55	55	65
4,000	60	80	80	80	90
6,000	70	100	100	100	110
8,000	120	180	180	180	190
10,000	220	290	290	290	300

TABLE II. Transmitter accuracy.

Test	Tolerance
Friction:	
Room Temperature	5 ppm
Low Temperature	5 ppm
High Temperature	5 ppm
Position Error	20 ppm
Voltage and Frequency Effect:	
Scale Error	20 ppm
Phase Angle	3°
Vibration:	
Variation	40 ppm
Oscillation	60 ppm (double amp)

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TABLE III. Phase angle tolerance.

Phase Angle Tolerance, Degree ±			
Flow PPM	Room Temp	Low Temp	High Temp
0	--	--	--
2,000	1.0	1.5	1.5
4,000	0.5	1.0	1.0
6,000	0.5	1.0	1.0
8,000	1.0	1.5	1.5
10,000	1.5	2.0	2.0

4.5.7 Low temperature scale error and friction. The transmitter shall be subjected to a temperature of approximately -54°C for a period of 4 hours. While still at this temperature, the transmitter shall be tested for scale error and friction as specified in 4.5.4. The scale error and the friction error shall not exceed the tolerances specified in Tables I, II, and III. The fuel use in this test shall be at a room temperature, and the ambient temperature around the transmitter shall be maintained at approximately -54°C.

4.5.8 Low temperature leak. The transmitter shall be subjected to a temperature of approximately -54°C for a period of 4 hours. While the transmitter and fuel are still at this temperature, the test specified in 4.5.3 shall be repeated. There shall be no leaks and the transmitter shall not be damaged as a result of this test. This test may be combined with the test specified in 4.5.7.

4.5.9 High temperature scale error and friction. The transmitter shall be subjected to a temperature of approximately 70°C for a period of 4 hours. While still at this temperature, the transmitter shall be tested for scale error and friction error as specified in 4.5.4. The scale error and friction error shall not exceed the tolerances in Tables I, II, and III. The fuel used in this test shall be at room temperature, and the ambient temperature around the transmitter shall be maintained at approximately 70°C.

4.5.10 Vibration resonances. For the vibration resonances tests, the transmitter shall be subjected to the following frequency range and vibratory acceleration or double amplitude:

Frequency (cpm)	Amplitude
300 - 600	0.010 inch
6,000 - 30,000	±5g
30,000 - 60,000	±5 g

4.5.10.1 When conducting the vibration resonance test and observing errors, the fuel flow rate shall be approximately 1,000 ppm. During this test, when the output signal error is not being observed, the vibration may be applied with zero flow but with fuel in the transmitter.

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4.5.10.2 The transmitter shall be electrically connected to a master voltmeter. The resonant frequency of the transmitter shall be determined by varying the frequency of applied vibration slowly through the range of frequencies and vibratory acceleration specified. The resonant frequency shall be determined by observing the output signal on the master voltmeter. The output signal vibration variation error and oscillation error shall be noted during the scanning of the frequency range for the resonant frequency. These vibration errors shall not exceed the tolerance specified in Table II.

4.5.10.3 The test specimen shall be vibrated at the indicated resonant condition for 1 hour (time specified refers to one axis of the vibration). During this period, the applied double amplitude or vibratory acceleration shall be as specified in 4.5.11.1. The periods of vibration shall be accomplished in sequence for each three mutually perpendicular axes vibration applied along any one axis. Each test period shall be conducted at the most severe resonance frequency.

4.5.10.4 When resonant frequencies are not apparent within the specified frequency range , the transmitter shall be vibrated twice as long as specified for resonance at a frequency of 3,000 cpm. At the end of this test, the transmitter shall be inspected to determine that no damage or defects have resulted due to this test. The transmitter shall then be subjected to the vibration cycling test specified in 4.5.11.

4.5.11 Vibration cycling. The transmitter shall be mounted and connected as specified in 4.5.10.1 and 4.5.10.2 and vibrated with the frequency cycling between 600 and 6,000 cpm in 15 minute cycles as the applied amplitude specified in 4.5.11.1. The vibration cycling shall be conducted for a period of 1 hour in each of the 3 mutually perpendicular axes. After the vibration test, a scale error test shall be conducted and the readings established shall not exceed the room temperature tolerances specified in Table I. The test specimen shall then be thoroughly examined, and there shall be no damage as a result of this test.

4.5.11.1 Vibration conditions during cycling shall be in accordance with the following:

Frequency (cpm)	Amplitude
300 - 600	0.050 inch
600 - 4,500	0.036 inch
4,500 - 30,000	±10 g
30,000 - 60,000	±5 g

4.5.12 Voltage and frequency variation effect. The applied voltage for this test shall be set at the following points:

a. 22 ½ V DC
   110 V AC
   380 Hertz

b. 24 V DC
   110 V AC
   420 Hertz

c. 24 V DC
   120 V AC
   380 Hertz

d. 30 V DC
   120 V AC
   420 Hertz

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4.5.12.1 When the applied voltages and frequencies are changed as specified, the errors in the output signal shall not exceed the room temperature tolerances specified in Tables I and III by more than the tolerance in Table II. Proper allowance shall be made for change of ac voltage from standard condition.

4.5.13 Power consumption. With nominal supply voltage applied to the transmitter, the power consumption shall not exceed the following maximum values:

4.5.13.1 The motor power shall not exceed 30 watts which shall be determined by measuring the DC voltage input to the master power supply and the dc amperes into the inverter of the master power supply. The watts shall be the product of volts and amperes.

4.5.13.2 The power consumed by the output signal circuit shall not exceed 7 VA when measured at the nominal test voltage.

4.5.14 Electrical characteristics. Sufficient electrical tests shall be performed by the manufacturer on the transmitter and its components to insure satisfactory performance when connected in a complete flowmeter system as specified in 3.2.

4.5.15 Radio noise. A radio noise test shall be performed with a frequency range of 0.15 to 150 MHz for radiated radio noise.

4.5.16 Pressure drop. The pressure drop of the transmitter shall be determined for the flow rates specified in Table IV and shall not exceed the values specified therein. The pressure drop values specified shall not be exceeded whether the transmitter is functioning or not functioning.

TABLE IV. Pressure drop flow rate.

Flow PPM	Max Pressure Drop PSI
2,000	0.25
4,000	1.0
6,000	2.3
8,000	4.0
10,000	6.25

4.5.17 Response time. The fuel flow shall be changed from 0 to 8,000 ppm by means of a rapid-action valve in the fuel line. The time for the indication of a master voltmeter connected to the transmitter to read within 1% of the steady state position shall not exceed 1 second.

4.5.18 Substitute fuel test. The transmitter shall be tested for output signal error with fuels in which the densities are approximately 10% above and 10% below, respectively, the calibrating fuel density used. The output signal errors shall not exceed the values in Table I.

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4.5.19 Fuel resistance test. The transmitter shall be tested for fuel resistance in accordance with SAE-ARP8615.

4.5.20 Humidity test. A humidity test shall be performed. Upon completion, the transmitter shall be subjected to the test specified in 4.5.4. The scale errors and friction errors shall not exceed the tolerances specified in TABLE I.

4.5.21 Fungus Resistance. A fungus resistance test shall be performed.

4.5.22 Salt spray. The transmitter shall be subjected to a salt spray test of 50 hours or longer according to ASTM-B117 Standard test Method of Salt Spray (Fog) Testing, using 5% salt solution. At the completion of this test, the transmitter shall be subjected to the test specified in 4.5.3 and 4.5.4. There shall be no leakage and the scale errors shall not exceed the tolerances specified in Table I.

4.5.23 Altitude test. An altitude test shall be conducted for a period of 4 hours for electrical components which are not hermetically sealed. Fuel flow will not be required during this test.

4.5.24 Explosion proof. A test shall be accomplished to ensure that the transmitter is explosion proof.

4.5.25 Shock. The transmitter shall be subjected to impact shocks of 15 g, each shock impulse having a time of 7 ±2 milliseconds. Three shocks shall be applied along the longitudinal axis and at 45° to the longitudinal axis in one direction. At the conclusion of this test, there shall be no evidence of damage, and the transmitter shall meet the room temperature tolerances.

4.5.26 Cycling operation. The system shall be operated continuously for 500 hours with actual or simulated flow through the transmitter. The actual or simulated cycling rate from zero to full scale and back to zero shall once each 2 ±1 minutes. No failure or any components shall occur. The accuracy of the transmitter shall be recorded before and after the test. The tolerances shall not exceed the values specified in Table I.

4.5.26.1 Simulated flow. The following procedure may be used in lieu of 500 hours of cycling with the actual flow:

a. The transmitter shall be operated for 50 hours with random flow variations of the type that are obtained in a flow stand when calibrating production transmitters.

b. The transmitter shall be disassembled by separating the turbine end from the impeller end.

c. The impeller shall be operated for 400 hours by exciting the drive motor in the normal manner. The impeller shall be immersed in fuel in the normal manner during operation.

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d. The turbine end shall be operated for 400 hours by applying a mechanical torque to the turbine causing it to deflect to full scale and back to zero at rate of once every 2 ±1 minutes. The turbine shall be immersed in fuel in the normal manner during this operation, and 115V 400 hertz power excitation shall be applied in the normal manner.

e. The transmitter shall be reassembled and operated for 50 hours with random flow variations of the type specified in “a” above.

4.5.27 Reverse-flow test. The transmitter shall be subjected to a rapidly applied reverse flow. This flow shall build up from 0 to a value of at least 60% of full scale in approximately 1 second. After 5 surges of reverse flow, the transmitter shall be retested in the normal manner and the scale error shall not exceed the room temperature tolerance specified in Table I.

4.5.28 Contaminated fuel operation. The transmitter shall be operated with contaminated fuel (80 grams of foreign matter per 1,000 gallons of fuel) at a flow rate of approximately 2,000 ppm for 1 hour. It shall then be operated with normal test fuel and a scale and friction error test conducted immediately. It shall operate satisfactorily and be within room temperature tolerances. The foreign matter shall consist of not less than 68% SiO_2 and shall have a particle size analysis as shown in Table V.

TABLE V. Contaminated fuel particle size.

Foreign Particles Size, Microns	Percent of Total Weight
0 - 5	39 ±2
5 - 10	18 ±3
10 - 20	16 ±3
20 – 40	18 ±3
over 40	9 ±3
Through a 200 mesh screen	100

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The MB-3 transmitter is intended for use I a fuel flowmeter system used for aircraft in-flight refueling operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Preservation, packaging and packing required (see 5.1).

6.3 Subject term (key word) listing.

Hermetic seal
JP-4
JP-5
JP-5/JP-8
Phase angle
Pounds per minute

6.4 Definitions.

6.4.1 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials or both. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

6.4.2 Phase angle. The phase angle is the electrical angle of input voltage and output voltage measured at the zero voltage point on the sine wave as the voltage goes from negative to positive.

6.4.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Air Force - 71

Preparing Activity:
Air Force - 71

(Project 6680-2013-001)

Review Activities:
Air Force - 99
DLA - GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T04:56Z
Check the source to verify that this is the current version before use.